



UNIVERSITY OF RUHUNA
BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY
ICT4163 - Digital Image Processing
Practical 03 – Image Digitization



1. Load and display your own analog image to your Google Collab environment.
2. Load and display your own digital image to your Google Collab environment.
3. Up sampling your analog image with the following up sampling scaling factors

Up sampling Factors:

- 1
- 2
- 4
- 8

4. Down sampling your analog image with the following down sampling scaling factors.

Down sampling Factors:

- 1
- 2
- 4
- 8

5. Follow the following steps and perform the quantization on your analog image.
 - a. Load your analog image to your Google Collab environment.
 - b. Define the following list of bins.

K1	2
K2	4
K3	8
K4	16
K5	32
K6	64

- c. Create bins for each k value with equally spaced intensity levels between 0 and maximum pixel value.
 - d. Assigns each pixel to the nearest bin.

- e. convert bin numbers back into intensity values.
 - f. Display results of each output image with respect to different bin variations.
6. Download any color image from the internet and perform the following conversions and display the output with the conversion you did.
- a. Convert the image to grayscale.
 - b. Convert the grayscale image to binary.
 - c. Convert the grayscale image to color.
 - d. Apply the jet colormap to grayscale image.